

Executive Summary:

This week we finished finalizing our plans for the product for CCEW. We created block diagrams for our individual portions of the circuits, software, and overall board. We also completed two meetings, one team meeting where we completed everything above, and the other with Mr.Nooner to combine questions with another team to see what else we needed to complete. We are finishing our PDR after starting it this week and it will have several charts created this week from every member.

Task / Time Take / Completed Y/N

Hardware:

Cody C.

Progress:

- 2/18 – Weekly meeting with the team (~1 hour)
 - Created a block diagram of the whole project
 - Created a rough budget for the project
 - Talked about using the 140 dB microphone and finding software solutions to detect shots vs general loud noises.
 - Talked about battery issues. Initially, we thought about using a circular battery, however, the current flow was not enough for the GPS module. We decided to use a lithium battery instead.
 - Throw around the idea of potentially using a DC-to-DC converter, to either have a boost converter or a buck converter
 - Finalized the general idea of which components to use
- Create Schematics symbol in KiCad for components (~1 Hour)

- I2S MEMS Microphone (~20 minutes)
- Button
- Audino Nano ESP32 Board
- NORA-B1?
- Created rough schematics of all components combined (~30 Minutes)
 - Research ways to combine I2S MEMS Microphone with the Arduino Nano ESP32 board.
 - Create a smaller version with a smaller chip?
- Research information for presentation (~30 Minutes)
- Finding potential software applications to use to create a block diagram (~30 Minutes)

Total Hours Worked: ~3 Hours 30 Minutes

Future Work/Plan:

- Create/Finalize the hardware block diagram
- Create presentation
- Practice presentation with the whole group

Task / Time Take / Completed Y/N

Hardware:

Ryan K.

Progress:

- This week I explored other positioning modules and decided to go with the Gravity: Positioning Module by DFRobot. I also researched some more to hopefully find a module

that offered the same features at a relatively lower current requirement, but was unsuccessful (2hrs). The selected module will allow high-precision and stable data for longitude, latitude, and altitude. Some research went into looking for a proper power supply (1hr). I also created a custom component in KiCad for the positioning module for future schematic and PCB designs (1hr).

Total Hours Worked: 4hrs

Future Work/Plan:

- Next week I will be working on the presentation for the PDR and creating a schematic for the device

Task / Time Take / Completed Y/N

Software:

Alex H.

Progress:

- 2/17 (1 h 23 min)
 - Researched additional processes needed to analyze and detect audio signals from gunshots. Found C++ library called arduinoFFT, which can compute the Fast Fourier Transform on a C array representing an audio signal in the time-domain. By converting the audio signal to the frequency domain using the Fast Fourier Transform, the audio magnitudes at different frequencies can be analyzed, and patterns for gunshots can be found (spectral analysis):

<https://github.com/kosme/arduinoFFT>

- Decided on sampling rate of at least 44.1 kHz, as 44.1 kHz is standard audio sampling rate.
- 2/18 (35 min)
 - **Team Meeting:** Discussed and finalized necessary functionality and components for the device as well as the feasibility of implementation. Software will need to utilize I2C communication protocol to obtain data from GPS module and I2S communication protocol to obtain data from microphone. Also discussed with the team a possible need for an SD card and subsequent SD card reader module to collect audio training data for machine learning model.
 - **Client Meeting:** Client emphasized the importance of creating a functional product over the aesthetic. Thus, making gunshot detection functional is a priority.
 - Read more intricate details of arduinoFFT library to further understand its features, including its window function tools.
- 2/20 (2 h 27 min)
 - Developed a block diagram for the software data flow. As a general overview for the audio signal processing, the microcontroller will first sample audio for a set amount of time and store the time-domain data in an array. Then, using the arduinoFFT library, a Hamming window function will be applied to the data to remove outliers. Then, the Fast Fourier Transform will be applied to the data to convert the audio sample into the frequency domain. Finally, using a tinyML model, spectral analysis will be performed to detect any gunshots in the data.
 - Added a hardware branch and software branch to the GitHub repository for this project.

- Created an empty Arduino project for the software and pushed it to the GitHub repository.

Total Hours Worked: (4 h 25 min)

Future Work/Plan:

- Continue working on PDR slides and practice presentation
- Finalize software block diagram and audio signal processing methodology for gunshot detection
- Install software library dependencies for the project

Project Manager:

Jonas W.

Progress:

- 2/17(45 Min)
 - Sent and Communicated with Dr. Fitzmorris and Mr. Nooner on scheduling meeting with other PM for weekly meeting
 - Made communication with other PM's and with Dr. Fitzmorris to finish scheduling for CDR, PDR, and Final
- 2/18(2 Hr)
 - Held a team meeting where we finalized the product features plan. We also finalized chips for the PCB we will be designing.
 - Held a meeting with other teams PM and Mr.Nooner, gave him the run down of our finalized plans and we had all our questions answered.
- 2/19 (1hr)
 - Created basic block diagram of circuit with basic data sending structure.

- Created the WAR and Presentation to send out to team to begin our PDR and finalize our WAR

Total Hours Worked: 3 Hr 45 Min

Future Work/Plan: Next week we are planning on finishing our PDR and doing practice runs on it for our presentation next Monday, 03/02. We will have a team meeting on Wednesday to complex our tables for the PDR. I will have the complete good looking Block diagram and hopefully our team will have the schematics as well fully completed.